built once from the fault-free period (Events: also daily)

1–2 SENSE

the building's own data, turned into events

Trend data

1

What the building-automation system already records: temperatures, flows, damper and valve positions, fan speeds, once a minute. No new sensors.

Configuration file

1

Maps sensor names to roles and lists the control sequence as rules. The only building-specific part: a new building means a new file, not code.

Events 2

one case per day, rebuilt every new day

State events: what the equipment is doing

Signature events: what should not happen

state events only

3–4a UNDERSTAND

what a normal day looks like, learned from one fault-free period

Process models

3

Discovered from the state events only, so they cannot learn the configured fault rules. One per terminal unit; one for the whole unit.

Calibration

4a

Healthy bands and model thresholds for every check come from the fault-free days. No threshold ever sees fault data.

Held-out days

4a

The last eight days of each month (96 a year) are kept aside and used only to measure each check's false-alarm rate.

today's events scored against the models

run on every new day

4b CHECK

eight checks score every new day, each against its own threshold

Absence	Unit conformance	Terminal-unit conformance	Frequency	Rate	Oscillation	Residual	Rules
expected terminal unit silent	day fits the unit model	day fits the terminal-unit model	daily count out of range	last 30 days against same month; support only	actuator reversing too often	physical relationship drifted	a control rule broken: a signature event

and go straight to the Rules check
signature events skip discovery

read the event log

read the raw sensors

each check's flagged days

5–6 ALERT

only what beats its own false-alarm rate reaches the operator

Noise gate

5

A check may alarm only when its flags exceed what its own false-alarm rate would produce by chance. The rate is never taken below 3 days in 96.

Alarm to the operator

6

What broke. Where: unit, terminal unit, zone.
The evidence: values and duration.
The check's measured false-alarm rate.

STRATA

a process-mining framework for HVAC fault detection